

# PE-EC802A — Mixed Signal Design

Semester VIII • 3L:0T:0P • 3 Credits

## Module 1 Exam Notes

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## 1. Analog vs. Discrete-Time Processing & Sampling Theory

### Analog vs. Discrete-Time Signals

An **analog signal** is continuous in both time and amplitude. A **discrete-time signal** is defined only at discrete time instants (continuous in amplitude, discrete in time). A **digital signal** is discrete in both time and amplitude (quantized).

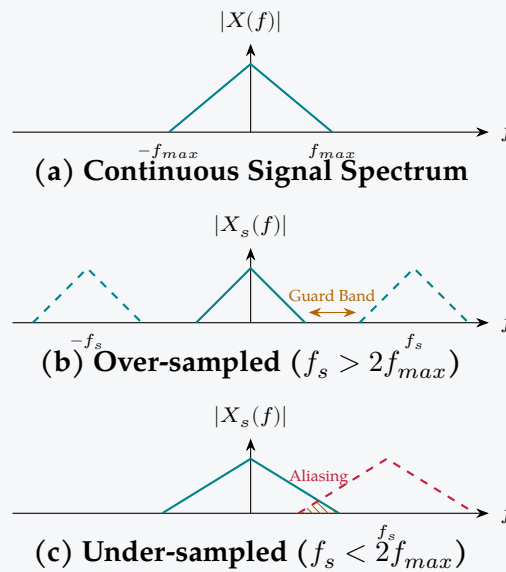
### 1.1 Sampling Theorem and Aliasing

To process an analog signal digitally, it must be sampled. The **Nyquist-Shannon Sampling Theorem** states that a band-limited continuous-time signal  $x_a(t)$  with maximum frequency  $f_{max}$  can be uniquely reconstructed from its samples  $x(n) = x_a(nT_s)$  if the sampling frequency  $f_s = 1/T_s$  satisfies:

$$f_s \geq 2f_{max} \quad (\text{Nyquist Rate})$$

If  $f_s < 2f_{max}$ , high-frequency components fold back into the lower frequency band, causing irreversible distortion known as **aliasing**. To prevent aliasing, an analog **Anti-Aliasing Filter (AAF)** (a low-pass filter) is placed before the sampler to limit the signal bandwidth to  $f_s/2$ .

### Sampling and Aliasing in the Frequency Domain



### 1.2 Signal Reconstruction

Reconstruction is the process of converting discrete samples back into a continuous-time signal. Mathematically, this is done by passing the impulse train through an ideal low-pass filter (cutoff  $f_c = f_s/2$ ). In time domain, this corresponds to the Whittaker-Shannon interpolation formula:

$$x_a(t) = \sum_{n=-\infty}^{\infty} x(n) \text{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

In practice, a non-ideal **Zero-Order Hold (ZOH)** is used, which introduces a frequency roll-off governed by the  $\text{sinc}(f/f_s)$  function, requiring a post-reconstruction smoothing filter.

## 2. Analog Continuous-Time Filters

### Butterworth vs. Chebyshev Filters

- **Butterworth Filter:** Characterized by a **maximally flat** magnitude response in the passband. The transfer function has only poles (no zeros). The magnitude squared is  $|H(j\omega)|^2 = 1/[1 + (\omega/\omega_c)^{2N}]$ .
- **Chebyshev Filter (Type I):** Permits **equiripple behavior** in the passband to achieve a sharper transition band (faster roll-off) for the same filter order  $N$ .

### 2.1 Passive RC and RL Filters

A passive filter uses only resistors, capacitors, and inductors. No power supply is needed, but no gain is possible. Passive filters are commonly used as anti-aliasing front-ends before ADC inputs.

#### First-Order Passive Filter Transfer Functions

**RC Low-Pass Filter** (series  $R$ , shunt  $C$ ):

$$H_{LP}(s) = \frac{1/sC}{R + 1/sC} = \frac{\omega_c}{s + \omega_c} \quad \omega_c = \frac{1}{RC}$$

**RC High-Pass Filter** (series  $C$ , shunt  $R$ ):

$$H_{HP}(s) = \frac{R}{R + 1/sC} = \frac{s}{s + \omega_c}$$

**RL Low-Pass Filter** (series  $R$ , shunt  $L$ ):

$$H_{LP}(s) = \frac{sL}{R + sL} \rightarrow \frac{\omega_c}{s + \omega_c} \quad \omega_c = \frac{R}{L}$$

Roll-off rate:  $-20$  dB/decade per filter order. At  $\omega = \omega_c$ :  $|H| = 1/\sqrt{2}$  ( $-3$  dB point).

#### Limitations of Passive Filters:

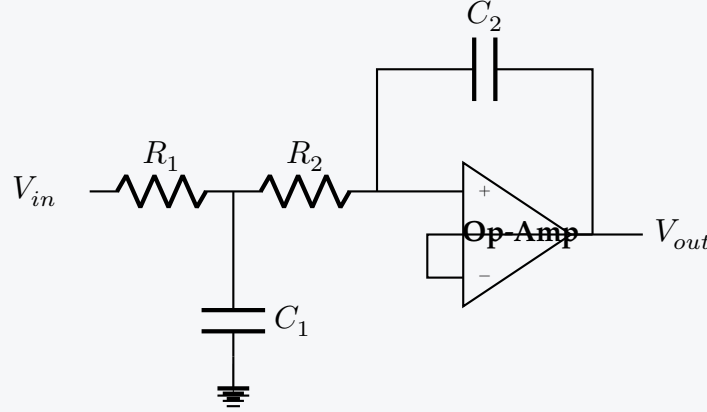
- **Insertion Loss:** Even in the passband, passive filters attenuate the signal (no gain possible).
- **Loading Effect:** Connecting a load changes the effective cutoff frequency unless a buffer is added.
- **Slow Roll-off:** Only  $-20$  dB/decade per order; higher-order cascaded passive stages are impractical.

Active filters (Sallen-Key, Tow-Thomas) solve these limitations using op-amps for isolation and gain.

## 2.2 Sallen-Key Active Biquad Filter

The Sallen-Key is a popular second-order active filter stage (biquad). It uses a single operational amplifier configured as a voltage follower (gain  $K \geq 1$ ) and two RC sections.

### Second-Order Sallen-Key Low-Pass Active Filter



For the Sallen-Key low-pass filter with  $R_1 = R_2 = R$  and  $C_1 = C_2 = C$  and unity amplifier gain ( $K = 1$ ):

### Sallen-Key Transfer Function and Parameters

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{s^2 R^2 C^2 + 2sRC + 1}$$

Comparing with the standard second-order biquad form  $H(s) = \frac{\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$ :

$$\omega_0 = \frac{1}{RC} \quad (\text{resonant angular frequency})$$

$$Q = \frac{1}{2} = 0.5 \quad (\text{quality factor})$$

## 2.3 Tow-Thomas State-Variable Biquad Filter

The Tow-Thomas biquad is a multi-op-amp active filter that is highly stable and easily tunable. It consists of an input summing integrator, a second integrator, and an inverting amplifier to complete the feedback loop.

The transfer function for the Tow-Thomas bandpass output is:

$$H_{BP}(s) = -\frac{s \frac{1}{R_4 C_1}}{s^2 + s \frac{1}{R_1 C_1} + \frac{1}{R_2 R_3 C_1 C_2}}$$

This topology allows independent tuning of the gain, frequency  $\omega_0$ , and quality factor  $Q$ .

## 2.4 Active Gm-C Filters

At high frequencies (where op-amp open-loop gain drops), active RC filters suffer. **Gm-C filters** replace resistors and op-amps with **Operational Transconductance Amplifiers (OTAs)** (which act as voltage-controlled current sources:  $I_{out} = g_m V_{in}$ ) and capacitors.

- **OTA Integrator:** Output current charges a capacitor:  $V_{out} = \frac{I_{out}}{sC} = \frac{g_m}{sC} V_{in}$ .
- **High-Frequency Tuning:** The transconductance  $g_m$  is electronically tunable by adjusting the OTA bias current, allowing real-time frequency tuning.

### 3. Basics of Analog Discrete-Time Filters & Z-Transform

#### Discrete-Time Filter Definition

A discrete-time filter processes sequence samples  $x(n)$  to produce output samples  $y(n)$ . It is described by a linear constant-coefficient difference equation:

$$y(n) = \sum_{i=0}^M b_i x(n-i) - \sum_{j=1}^N a_j y(n-j)$$

#### 3.1 Z-Transform and ROC

The Z-transform is the discrete-time equivalent of the Laplace transform. For a sequence  $x(n)$ , it is defined as:

$$X(z) = \mathcal{Z}\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

where  $z = re^{j\omega}$  is a complex variable. The **Region of Convergence (ROC)** is the set of values in the  $z$ -plane for which the summation converges.

- **Causal sequence:** ROC is the exterior of a circle ( $|z| > R_{max}$ ).
- **Stable system:** The ROC must contain the unit circle ( $|z| = 1$ ). This implies all poles of the system transfer function  $H(z)$  must lie strictly inside the unit circle ( $|poles| < 1$ ).

#### 3.2 Mapping s-Plane to z-Plane

To design discrete-time filters from continuous-time templates, the  $s$ -plane must be mapped to the  $z$ -plane.

1. **Impulse Invariance:** Preserves the impulse response shape. The mapping is  $z = e^{sT_s}$ . This suffers from severe spectral aliasing.
2. **Bilinear Transformation (BLT):** Maps the entire left half of the  $s$ -plane to the inside of the unit circle in the  $z$ -plane. It is defined by the algebraic mapping:

$$s = \frac{2}{T_s} \frac{z-1}{z+1} \quad \Rightarrow \quad z = \frac{1 + sT_s/2}{1 - sT_s/2}$$

#### 3.3 Frequency Warping Effect

The BLT maps the infinite continuous frequency axis  $-\infty < \Omega < \infty$  non-linearly to the finite discrete frequency axis  $-\pi/T_s < \omega < \pi/T_s$ . Substituting  $s = j\Omega$  and  $z = e^{j\omega T_s}$  gives the frequency relation:

## Frequency Warping and Pre-warping Relations

$$\Omega = \frac{2}{T_s} \tan\left(\frac{\omega T_s}{2}\right)$$

To compensate for this non-linear compression (warping), continuous-time specifications must be **pre-warped** before applying the BLT:

$$\Omega_{pre} = \frac{2}{T_s} \tan\left(\frac{\omega_{desired} T_s}{2}\right)$$

## Worked Example: Bilinear Transformation with Pre-warping

**Problem:** Design a discrete low-pass filter with cutoff  $f_d = 2$  kHz at sampling rate  $f_s = 10$  kHz. Use a first-order continuous-time prototype  $H(s) = \frac{\Omega_c}{s + \Omega_c}$ .

**Solution:**

- Calculate  $T_s = 1/f_s = 10^{-4}$  s.
- Discrete angular frequency  $\omega_d = 2\pi f_d = 4000\pi$  rad/s.
- Pre-warped continuous frequency  $\Omega_c$ :

$$\Omega_c = \frac{2}{10^{-4}} \tan\left(\frac{4000\pi \times 10^{-4}}{2}\right) = 20000 \tan(0.2\pi) \approx 20000 \times 0.7265 = 14530 \text{ rad/s}$$

- Apply Bilinear Transformation:

$$H(z) = \frac{\Omega_c}{s + \Omega_c} \bigg|_{s = \frac{2}{T_s} \frac{z-1}{z+1}} = \frac{14530}{\frac{2}{10^{-4}} \frac{z-1}{z+1} + 14530} = \frac{14530(z+1)}{20000(z-1) + 14530(z+1)}$$

$$H(z) \approx \frac{14530(z+1)}{34530z - 5470} \approx \frac{0.42(z+1)}{z - 0.158}$$

## Quick Revision — Module 1

Filter Fundamentals and Z-Transform | PE-EC802A

### Key Formulas and Transforms at a Glance

|                                   |                                                                                                                          |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Nyquist Sampling Rate</b>      | $f_s \geq 2f_{max}$                                                                                                      |
| <b>Sinc Reconstruction</b>        | $x_a(t) = \sum x(n) \text{sinc}\left(\frac{t-nT_s}{T_s}\right)$                                                          |
| <b>Sallen-Key LP frequency</b>    | $\omega_0 = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}; \quad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{R_1 C_2 + R_2 C_2 + (1-K)R_1 C_1}$ |
| <b>Bilinear Transformation</b>    | $s = \frac{2}{T_s} \frac{z-1}{z+1}$                                                                                      |
| <b>Frequency Warping relation</b> | $\Omega = \frac{2}{T_s} \tan\left(\frac{\omega T_s}{2}\right)$                                                           |
| <b>Z-transform</b>                | $X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$                                                                            |
| <b>Unit circle frequency map</b>  | $z = e^{j\omega T_s}$                                                                                                    |

### One-Line Definitions

- **Aliasing:** High frequencies folding back to lower bands due to under-sampling ( $f_s < 2f_{max}$ ).
- **Anti-Aliasing Filter:** A low-pass filter limiting analog signal bandwidth to  $f_s/2$  before sampling.
- **Butterworth Response:** Maximally flat magnitude profile in the passband.
- **Chebyshev Response:** Passband equiripple profile resulting in a sharper cutoff transition band.
- **Biquad Filter:** A second-order active filter block implementing a pair of poles and optionally zeros.
- **Gm-C Filter:** A high-frequency filter using transconductance amplifiers and capacitors.
- **ROC:** The region in the z-plane where the Z-transform power series converges.
- **Frequency Warping:** Nonlinear compression of the continuous frequency axis introduced by the bilinear transformation.

## Analog Filter Topology Comparison

| Type          | Active Elements | Key Property             | Use Case               |
|---------------|-----------------|--------------------------|------------------------|
| Passive RC/RL | None            | No gain, simple          | AAF, RF matching       |
| Butterworth   | Op-amp          | Maximally flat passband  | General purpose LP     |
| Chebyshev     | Op-amp          | Equiripple, sharp cutoff | Steeper rolloff needed |
| Sallen-Key    | 1 op-amp        | 2nd-order, easy design   | Active LP/HP/BP        |
| Tow-Thomas    | 3 op-amps       | Tunable $\omega_0, Q$    | Precision biquad       |
| Gm-C          | OTA (no R)      | High-freq, tunable       | > 100 MHz filters      |

## Frequently Asked / Viva Questions

**Q1. State the sampling theorem. What happens if the sampling rate is below the Nyquist rate?**

The sampling theorem states that the sampling rate  $f_s$  must be at least twice the maximum signal frequency component ( $f_s \geq 2f_{max}$ ). Under-sampling causes high frequencies to overlap with low frequencies, creating irreversible aliasing distortion.

**Q2. What is the function of the anti-aliasing filter?**

It is an analog low-pass filter placed before the sampler to block all frequency components above  $f_s/2$ , ensuring the sampling condition is satisfied.

**Q3. Compare Butterworth and Chebyshev filter characteristics.**

Butterworth filters are maximally flat in the passband but have a slower roll-off. Chebyshev filters have equiripple passband characteristics and a much steeper roll-off transition band.

**Q4. Explain the Sallen-Key active filter configuration.**

It is a second-order active filter structure using one operational amplifier configured as a voltage follower, providing high input impedance, low output impedance, and easily calculable components.

**Q5. What are the advantages of Gm-C filters over active RC filters?**

Gm-C filters do not need operational amplifiers or resistors. They use transconductance amplifiers (OTAs) and capacitors, making them suitable for high-frequency operation (up to hundreds of MHz) and electronically tunable.

**Q6. Why does the bilinear transformation map the left half of the s-plane to the inside of the z-plane unit circle?**

The BLT maps  $s = \sigma + j\Omega$  using  $z = (1 + sT/2)/(1 - sT/2)$ . For  $\sigma < 0$  (LHP), the numerator magnitude is smaller than the denominator magnitude, ensuring  $|z| < 1$  (inside the unit circle).

**Frequently Asked / Viva Questions (contd.)**

**Q7. What is frequency warping in the Bilinear Transformation, and how is it compensated?**

It is the non-linear compression of the continuous frequency axis  $\Omega$  to the discrete axis  $\omega$ . It is compensated by pre-warping continuous-time critical frequencies:  $\Omega_c = \frac{2}{T_s} \tan(\omega_d T_s / 2)$ .

**Q8. State the condition for a discrete-time LTI system to be stable.**

The system is stable if its Z-transform region of convergence (ROC) contains the unit circle ( $|z| = 1$ ), which requires all poles to lie strictly inside the unit circle.

**Q9. What is the Z-transform of a causal sequence  $a^n u(n)$ ? State its ROC.**

The Z-transform is  $H(z) = \frac{z}{z-a}$  (or  $\frac{1}{1-az^{-1}}$ ) with the region of convergence  $|z| > |a|$ .

**Q10. Why is the impulse invariance mapping not suitable for high-pass and band-stop filters?**

Impulse invariance maps frequencies linearly, which causes severe aliasing at high frequencies. High-pass and band-stop filters have infinite high-frequency energy, making aliasing unacceptable.